

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460673.8819 +/- 0.0058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.048
Transit depth (Rp/Rs)^2	1.93 +/- 1.34 [%]
Orbital Inclination (inc)	87.44 +/- 2.85
Airmass coefficient 1 (a1)	0.0133 +/- 0.0081
Airmass coefficient 2 (a2)	2.96 +/- 1.4
Scatter in the residuals of the lightcurve fit is	60.84 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	6.5
Transit Duration (day)	0.0855 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.048 vs 0.1368 (deviation 0.04σ)
Duration fit vs published	0.0855 d vs 0.0758 d (deviation 0.95σ)
Mid-time O–C	-14.1 min
Depth SNR	2.7
Residual scatter	60.84 %

Light curve

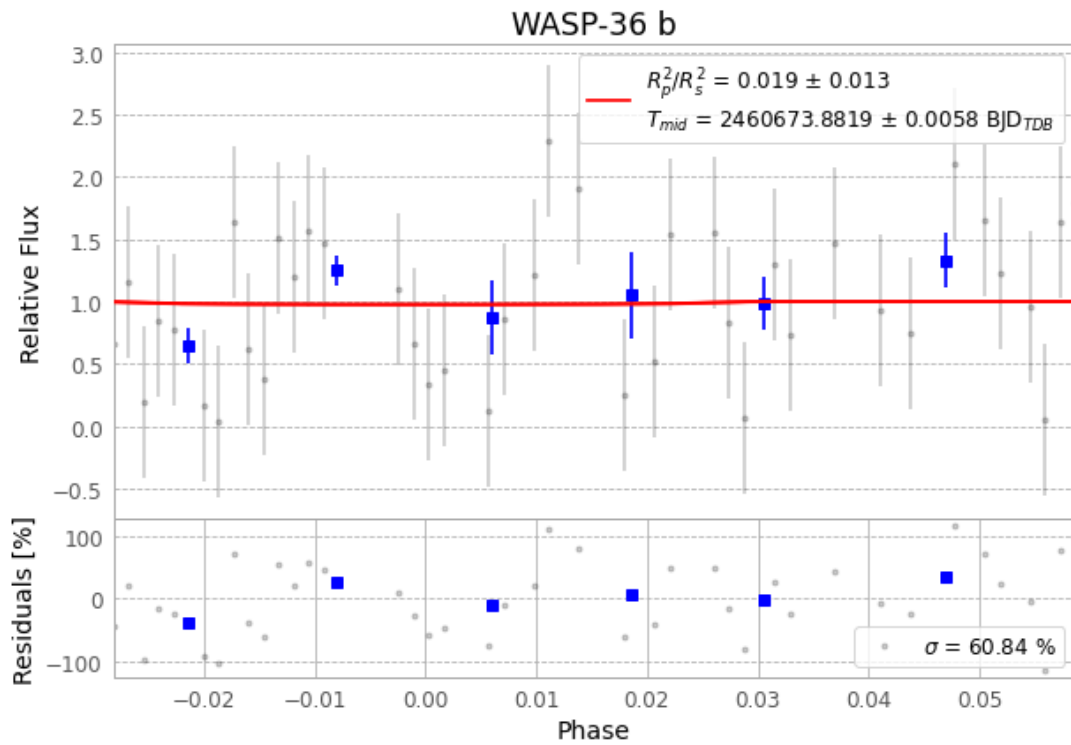


Figure 1 — FinalLightCurve_WASP-36 b_2024-12-29.png (SHA-256 e99603648725453a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [596, 390], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b2e4838c63a34118...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

65 FITS frames (43 MB), WASP-36241229080016.FITS → WASP-36241229111216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-241229.log (74182badd2f5...), FinalLightCurve_WASP-36 b_2024-12-29.png (e99603648725...), FinalLightCurve_WASP-36 b_2024-12-29.csv (f59de419ca44...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:30Z.